

Initial Parameter Configuration.

- CONTEXT_LENGTH: 30
- D_MODEL: 32
- *_LAYERS: 2
- *_HEADS: 4
- DROPOUT: 0.1
- LEARNING_RATE: 1e-4, 5e-5
- BATCH_SIZE: 32, 64, 128
- EPOCHS: 20 (with PATIENCE=10)

Batch Size 64, Learning Rate 1e-4

Epoch 01/20 -> Train Loss: -1.26991204, Val Loss: -1.65727360
New best validation loss: -1.65727360. Saving model...

Epoch 02/20 -> Train Loss: -2.20737763, Val Loss: -2.21042946
New best validation loss: -2.21042946. Saving model...

Epoch 03/20 -> Train Loss: -2.50152380, Val Loss: -2.42937358
New best validation loss: -2.42937358. Saving model...

Epoch 04/20 -> Train Loss: -2.64044149, Val Loss: -2.35278372
Validation loss did not improve. Patience: 1/10

Epoch 05/20 -> Train Loss: -2.73635860, Val Loss: -2.59325069
New best validation loss: -2.59325069. Saving model...

Epoch 06/20 -> Train Loss: -2.79875934, Val Loss: -2.73347369
New best validation loss: -2.73347369. Saving model...

Epoch 07/20 -> Train Loss: -2.83933081, Val Loss: -2.61346611
Validation loss did not improve. Patience: 1/10

Epoch 08/20 -> Train Loss: -2.88090314, Val Loss: -2.80592543
New best validation loss: -2.80592543. Saving model...

Epoch 09/20 -> Train Loss: -2.90326997, Val Loss: -2.74745058
Validation loss did not improve. Patience: 1/10

Epoch 10/20 -> Train Loss: -2.93674289, Val Loss: -2.73024203
Validation loss did not improve. Patience: 2/10

Epoch 11/20 -> Train Loss: -2.95575961, Val Loss: -2.88569784
New best validation loss: -2.88569784. Saving model...

Epoch 12/20 -> Train Loss: -2.97721556, Val Loss: -2.72488405
Validation loss did not improve. Patience: 1/10

Epoch 13/20 -> Train Loss: -2.99184759, Val Loss: -2.90419203
New best validation loss: -2.90419203. Saving model...

Epoch 14/20 -> Train Loss: -3.01072159, Val Loss: -2.88882065
Validation loss did not improve. Patience: 1/10

Epoch 15/20 -> Train Loss: -3.02630497, Val Loss: -2.94505787
New best validation loss: -2.94505787. Saving model...

Epoch 16/20 -> Train Loss: -3.05087575, Val Loss: -2.88568141
Validation loss did not improve. Patience: 1/10

Epoch 17/20 -> Train Loss: -3.05573125, Val Loss: -2.92449716
Validation loss did not improve. Patience: 2/10

Epoch 18/20 -> Train Loss: -3.06843116, Val Loss: -2.97391046
New best validation loss: -2.97391046. Saving model...

Epoch 19/20 -> Train Loss: -3.07660289, Val Loss: -3.06566454
New best validation loss: -3.06566454. Saving model...

Epoch 20/20 -> Train Loss: -3.08992422, Val Loss: -2.98136848
Validation loss did not improve. Patience: 1/10

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Final Transformer Model Performance

Dataset Metric Value (EURUSD)

Test RMSE 0.000320

Test MAE 0.000207

Validation RMSE 0.000405

Validation MAE 0.000277

Training RMSE 0.000400

Training MAE 0.000268

Batch Size 32, Learning Rate 1e-4

Smaller batch, same LR

Epoch 01/20 -> Train Loss: -1.60333151, Val Loss: -2.04360239
New best validation loss: -2.04360239. Saving model...

Epoch 02/20 -> Train Loss: -2.47938321, Val Loss: -2.42506824
New best validation loss: -2.42506824. Saving model...

Epoch 03/20 -> Train Loss: -2.68585628, Val Loss: -2.56477364
New best validation loss: -2.56477364. Saving model...

Epoch 04/20 -> Train Loss: -2.79625795, Val Loss: -2.66483450
New best validation loss: -2.66483450. Saving model...

Epoch 05/20 -> Train Loss: -2.85636547, Val Loss: -2.69993812
New best validation loss: -2.69993812. Saving model...

Epoch 06/20 -> Train Loss: -2.91095833, Val Loss: -2.74879452
New best validation loss: -2.74879452. Saving model...

Epoch 07/20 -> Train Loss: -2.94372955, Val Loss: -2.92939021
New best validation loss: -2.92939021. Saving model...

Epoch 08/20 -> Train Loss: -2.97415407, Val Loss: -2.88674768
Validation loss did not improve. Patience: 1/10

Epoch 09/20 -> Train Loss: -3.00215138, Val Loss: -2.96929288
New best validation loss: -2.96929288. Saving model...

Epoch 10/20 -> Train Loss: -3.01880060, Val Loss: -3.01218235
New best validation loss: -3.01218235. Saving model...

Epoch 11/20 -> Train Loss: -3.04151031, Val Loss: -2.90215657
Validation loss did not improve. Patience: 1/10

Epoch 12/20 -> Train Loss: -3.05396688, Val Loss: -2.93568177
Validation loss did not improve. Patience: 2/10

Epoch 13/20 -> Train Loss: -3.07187442, Val Loss: -3.01766368
New best validation loss: -3.01766368. Saving model...

Epoch 14/20 -> Train Loss: -3.08045147, Val Loss: -2.96974381
Validation loss did not improve. Patience: 1/10

Epoch 15/20 -> Train Loss: -3.09243840, Val Loss: -2.75313821
Validation loss did not improve. Patience: 2/10

Epoch 16/20 -> Train Loss: -3.10334807, Val Loss: -2.35868675
Validation loss did not improve. Patience: 3/10

Epoch 17/20 -> Train Loss: -3.10975877, Val Loss: -3.05532936
New best validation loss: -3.05532936. Saving model...

Epoch 18/20 -> Train Loss: -3.12512646, Val Loss: -3.11970832
New best validation loss: -3.11970832. Saving model...

Epoch 19/20 -> Train Loss: -3.12685709, Val Loss: -3.07627640
Validation loss did not improve. Patience: 1/10

Epoch 20/20 -> Train Loss: -3.14116334, Val Loss: -3.05601124
Validation loss did not improve. Patience: 2/10

Final Transformer Model Performance			
Dataset	Metric	Value	(EURUSD)
Test	RMSE	0.000328	
Test	MAE	0.000214	
Validation	RMSE	0.000422	
Validation	MAE	0.000287	
Training	RMSE	0.000383	
Training	MAE	0.000253	

Batch Size 128, Learning Rate 1e-4

Larger batch, same LR

Epoch 01/20 -> Train Loss: -0.92686727, Val Loss: -1.23884412
New best validation loss: -1.23884412. Saving model...

Epoch 02/20 -> Train Loss: -1.84993386, Val Loss: -1.83560488
New best validation loss: -1.83560488. Saving model...

Epoch 03/20 -> Train Loss: -2.24707492, Val Loss: -2.08755592
New best validation loss: -2.08755592. Saving model...

Epoch 04/20 -> Train Loss: -2.43074470, Val Loss: -2.25870916
New best validation loss: -2.25870916. Saving model...

Epoch 05/20 -> Train Loss: -2.54868340, Val Loss: -2.34327626
New best validation loss: -2.34327626. Saving model...

Epoch 06/20 -> Train Loss: -2.63521180, Val Loss: -2.44971000
New best validation loss: -2.44971000. Saving model...

Epoch 07/20 -> Train Loss: -2.70162389, Val Loss: -2.60593627
New best validation loss: -2.60593627. Saving model...

Epoch 08/20 -> Train Loss: -2.76107950, Val Loss: -2.65182109
New best validation loss: -2.65182109. Saving model...

Epoch 09/20 -> Train Loss: -2.78936203, Val Loss: -2.71789335
New best validation loss: -2.71789335. Saving model...

Epoch 10/20 -> Train Loss: -2.81347702, Val Loss: -2.75657008
New best validation loss: -2.75657008. Saving model...

Epoch 11/20 -> Train Loss: -2.84738540, Val Loss: -2.77831866
New best validation loss: -2.77831866. Saving model...

Epoch 12/20 -> Train Loss: -2.87400607, Val Loss: -2.77616874
Validation loss did not improve. Patience: 1/10

Epoch 13/20 -> Train Loss: -2.89928596, Val Loss: -2.83426776
New best validation loss: -2.83426776. Saving model...

Epoch 14/20 -> Train Loss: -2.92268465, Val Loss: -2.79895549
Validation loss did not improve. Patience: 1/10

Epoch 15/20 -> Train Loss: -2.94889859, Val Loss: -2.88064457

New best validation loss: -2.88064457. Saving model...

Epoch 16/20 -> Train Loss: -2.95165556, Val Loss: -2.84667892
Validation loss did not improve. Patience: 1/10

Epoch 17/20 -> Train Loss: -2.95941586, Val Loss: -2.87526496
Validation loss did not improve. Patience: 2/10

Epoch 18/20 -> Train Loss: -2.98167769, Val Loss: -2.89621100
New best validation loss: -2.89621100. Saving model...

Epoch 19/20 -> Train Loss: -2.98257849, Val Loss: -2.67427485
Validation loss did not improve. Patience: 1/10

Epoch 20/20 -> Train Loss: -2.99552825, Val Loss: -2.91034469
New best validation loss: -2.91034469

Final Transformer Model Performance		
Dataset	Metric	Value (EURUSD)
Test	RMSE	0.000506
Test	MAE	0.000410
Validation	RMSE	0.000646
Validation	MAE	0.000476
Training	RMSE	0.000436
Training	MAE	0.000309

Batch Size 64, Learning Rate 5e-5

Baseline batch, smaller LR

Epoch 01/20 -> Train Loss: -0.82771357, Val Loss: -0.95551336
New best validation loss: -0.95551336. Saving model...

Epoch 02/20 -> Train Loss: -1.58929005, Val Loss: -1.61279013
New best validation loss: -1.61279013. Saving model...

Epoch 03/20 -> Train Loss: -2.09533821, Val Loss: -2.00938993
New best validation loss: -2.00938993. Saving model...

Epoch 04/20 -> Train Loss: -2.32212109, Val Loss: -2.20630115
New best validation loss: -2.20630115. Saving model...

Epoch 05/20 -> Train Loss: -2.45695836, Val Loss: -2.35099105
New best validation loss: -2.35099105. Saving model...

Epoch 06/20 -> Train Loss: -2.55813019, Val Loss: -2.41197233
New best validation loss: -2.41197233. Saving model...

Epoch 07/20 -> Train Loss: -2.63158181, Val Loss: -2.46218513
New best validation loss: -2.46218513. Saving model...

Epoch 08/20 -> Train Loss: -2.68984730, Val Loss: -2.58541450
New best validation loss: -2.58541450. Saving model...

Epoch 09/20 -> Train Loss: -2.73715954, Val Loss: -2.57480919
Validation loss did not improve. Patience: 1/10

Epoch 10/20 -> Train Loss: -2.77197713, Val Loss: -2.69875941
New best validation loss: -2.69875941. Saving model...

Epoch 11/20 -> Train Loss: -2.81595377, Val Loss: -2.53935671
Validation loss did not improve. Patience: 1/10

Epoch 12/20 -> Train Loss: -2.85041870, Val Loss: -2.66764580
Validation loss did not improve. Patience: 2/10

Epoch 13/20 -> Train Loss: -2.87323641, Val Loss: -2.80376592
New best validation loss: -2.80376592. Saving model...

Epoch 14/20 -> Train Loss: -2.88469367, Val Loss: -2.76384907
Validation loss did not improve. Patience: 1/10

Epoch 15/20 -> Train Loss: -2.91203183, Val Loss: -2.77686467

Validation loss did not improve. Patience: 2/10

Epoch 16/20 -> Train Loss: -2.93391655, Val Loss: -2.80701717
New best validation loss: -2.80701717. Saving model...

Epoch 17/20 -> Train Loss: -2.94629628, Val Loss: -2.82033894
New best validation loss: -2.82033894. Saving model...

Epoch 18/20 -> Train Loss: -2.96192544, Val Loss: -2.85020253
New best validation loss: -2.85020253. Saving model...

Epoch 19/20 -> Train Loss: -2.97722806, Val Loss: -2.91205437
New best validation loss: -2.91205437. Saving model...

Epoch 20/20 -> Train Loss: -2.98369004, Val Loss: -2.95051577
New best validation loss: -2.95051577. Saving model...

Final Transformer Model Performance			
Dataset	Metric	Value	(EURUSD)
Test	RMSE	0.000322	
Test	MAE	0.000209	
Validation	RMSE	0.000422	
Validation	MAE	0.000291	
Training	RMSE	0.000423	
Training	MAE	0.000285	

Batch Size 32, Learning Rate 5e-5

Smaller batch, smaller LR

Epoch 01/20 -> Train Loss: -1.08562695, Val Loss: -1.33876782
New best validation loss: -1.33876782. Saving model...

Epoch 02/20 -> Train Loss: -2.02204218, Val Loss: -2.03741201
New best validation loss: -2.03741201. Saving model...

Epoch 03/20 -> Train Loss: -2.38029182, Val Loss: -2.32438763
New best validation loss: -2.32438763. Saving model...

Epoch 04/20 -> Train Loss: -2.55532383, Val Loss: -2.33318568
New best validation loss: -2.33318568. Saving model...

Epoch 05/20 -> Train Loss: -2.65354580, Val Loss: -2.56226696
New best validation loss: -2.56226696. Saving model...

Epoch 06/20 -> Train Loss: -2.72834142, Val Loss: -2.64622490
New best validation loss: -2.64622490. Saving model...

Epoch 07/20 -> Train Loss: -2.78032062, Val Loss: -2.60566478
Validation loss did not improve. Patience: 1/10

Epoch 08/20 -> Train Loss: -2.82453976, Val Loss: -2.74756510
New best validation loss: -2.74756510. Saving model...

Epoch 09/20 -> Train Loss: -2.85475230, Val Loss: -2.73367696
Validation loss did not improve. Patience: 1/10

Epoch 10/20 -> Train Loss: -2.88269752, Val Loss: -2.67491769
Validation loss did not improve. Patience: 2/10

Epoch 11/20 -> Train Loss: -2.91273476, Val Loss: -2.83435038
New best validation loss: -2.83435038. Saving model...

Epoch 12/20 -> Train Loss: -2.93159293, Val Loss: -2.82450196
Validation loss did not improve. Patience: 1/10

Epoch 13/20 -> Train Loss: -2.95163402, Val Loss: -2.91171420
New best validation loss: -2.91171420. Saving model...

Epoch 14/20 -> Train Loss: -2.97201619, Val Loss: -2.83493787
Validation loss did not improve. Patience: 1/10

Epoch 15/20 -> Train Loss: -2.99422154, Val Loss: -2.80398476
Validation loss did not improve. Patience: 2/10

Epoch 16/20 -> Train Loss: -3.00391598, Val Loss: -2.96117007
New best validation loss: -2.96117007. Saving model...

Epoch 17/20 -> Train Loss: -3.01906780, Val Loss: -2.91488574
Validation loss did not improve. Patience: 1/10

Epoch 18/20 -> Train Loss: -3.03616351, Val Loss: -3.01448142
New best validation loss: -3.01448142. Saving model...

Epoch 19/20 -> Train Loss: -3.04380723, Val Loss: -2.95063041
Validation loss did not improve. Patience: 1/10

Epoch 20/20 -> Train Loss: -3.06005004, Val Loss: -3.01103572
Validation loss did not improve. Patience: 2/10

Final Transformer Model Performance		
Dataset	Metric	Value (EURUSD)
Test	RMSE	0.000374
Test	MAE	0.000264
Validation	RMSE	0.000518
Validation	MAE	0.000354
Training	RMSE	0.000383
Training	MAE	0.000251